MUHAMMAD UMAR H

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WEBSITES, PORTFOLIOS, PROFILES

GitHub: https://github.com/Mhd-Umar

Portfolio: https://mhd-umar.github.io/Mhd-Umar_portfolio.github.io/

LinkedIn: https://www.linkedin.com/in/muhammad-umar-h-90641034a/

PROFESSIONAL SUMMARY

Entry-level Software Developer and Computer Hardware Engineering diploma student (CGPA 9.71) with hands-on experience in web development, databases and machine learning. Built a Django web application for laboratory management with a team and is developing an AI-based ransomware detection system for Windows. Certified AI/ML Telecom Data Analyst (NSDC, Skill India). Seeking a Software Developer, Backend, Frontend or IoT role.

SKILLS

- **Programming Languages:** Python, C, Java, JavaScript, PHP.
- **Database:** MySQL, SQL, SQLite, Database Management (intermediate).
- **Networking:** Networking fundamentals (intermediate).
- **IoT and Systems:** Computer systems and IoT fundamentals.
- **Frameworks and Tools:** Django, HTML, CSS, scikit-learn, VS Code.
- **AI/ML and Data:** Data Science and Data Analysis, NumPy, Machine Learning (ML), Artificial Intelligence (AI) (intermediate).
- **Core Competencies:** Backend Development (intermediate), Frontend Development, Software Development, Cybersecurity Awareness, Problem-Solving.

EXPERIENCE

Fresher with hands-on project experience in web development, databases and machine learning.

ResoTrack – College Laboratory Resource Analyzer and Management System | 2026 | Python, Django, HTML, CSS, JavaScript, SQLite.

- Developed a Django web application with 3 role-based modules (student, lab administrator, main admin) to manage laboratory resources, stock, complaints and suggestions across 4 engineering departments.
- Built a resource analyzer that shows department-wise and lab-wise resources with working, damaged and under-maintenance status.
- Enabled lab administrators to add, edit and delete resources, track 4 stock measures (quantity, available, damaged, received) and generate reports.
- Implemented student registration, login, complaint filing and status tracking, with admin approval of complaints and suggestions.
- Created a main admin control center to manage students and lab admins, create labs, assign lab admins and monitor activity.
- Collaborated with a team to deliver 16 screens across public, student, lab admin and main admin views, from database to user interface.

AI-Based Ransomware Early Detection and Warning System (Windows) | 2026 (In Progress) | Python, NumPy, Machine Learning, Isolation Forest.

- Designed a behavior-based detector that identifies ransomware during encryption using 5 signals: file change speed, entropy jumps, extension changes, parent process and shadow copy deletion attempts.
- Modeled an AI risk score (0-100) using Isolation Forest and One-Class SVM, with SHAP explanations for every alert.
- Programmed real-time file monitoring in Python with entropy checks and honeyfiles to catch encryption early.
- Engineered an always-on-top alert with kill, investigate or ignore options, process suspension and a live dashboard that runs fully on one PC.

EDUCATION

Government Polytechnic College, Attingal, Kerala.
Diploma, Computer Hardware Engineering (Lateral Entry), 2027.
GPA: 9.71 (Semester 3 SGPA: 9.41, Semester 4 SGPA: 10.0).

Higher Secondary (Plus Two), 2024.
GPA: 78.67%.

Secondary School (SSLC / 10th), 2022.
GPA: Full A+.

CERTIFICATIONS

- AI/ML Telecom Data Analyst - Certified by NSDC (Skill India) through the Board of Vocational Higher Secondary Examinations, Kerala.
- Hackathon 2026 - Participated up to district level in a hackathon by IEDC and Kerala Startup Mission.